Distance between two points

1

a $AB = 3$ units **b** $AB = 5$ units **c** $AB = 10$ units **d** $AB = 5$ units

2

a $AB = 6$ units **b** $AB = 3$ units **c** $AB = 14$ units **d** $AB = 8$ units

3

a $AC = 5$ units **b** $AC = 3.61$ units

4

a

i $PQ = 3$ units **ii** $QR = 4$ units **iii** $PR = 5$

b

i $PQ = 3$ units **ii** $QR = 4$ units **iii** $PR = 5$

c

i $PQ = 4$ units **ii** $QR = 7$ units **iii** $PR = 8.06$

d

i $PQ = 5$ units **ii** $QR = 4$ units **iii** $PR = 6.40$

5

a

i $AB = 5$ units **ii** $BC = 12$ units

iii $c = 13$

b

i $AB = 11$ units **ii** $BC = 7$ units

iii $c = 13.0$

6 11.31 units

Distance formula

7

a 20 **b** 20 **c** 13.04 **d** 6.40

8

a $AC = 5.39$ **b** $AC = 5$ **c** $AC = 6.40$ **d** $AC = 7.81$

e $AC = 11.40$ **f** $AC = 17$

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9

a 10 units b 15 units c $\sqrt{73}$ units d $\sqrt{34}$units

10

a $\sqrt{82}$ units b $\sqrt{26}$ units c N

11

a $AB = 9.90$ units b $PN = 11.18$ units

c N

12

a $AB = 3.61$ units b $AM = 1.80$ units

13 $M\,(-2, 9)$ and $M\,(-2, 3)$

14

a $AC = \sqrt{13}$ b $BC = \sqrt{13}$

c C is equidistant from A and B.

15 $AK = 7.2$ units

Geometric applications

16

a $BC = 6$ b $AB = \sqrt{18}$ c $AC = \sqrt{18}$ d Isosceles

17

a

i $AC = \sqrt{200}$ units ii $BD = \sqrt{72}$units

b 60 units^2

18

a 17 units

b 34π units

c 17 units

d Yes, because the distance between the centre and the point $(20, 4)$ is the same as the radius.

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19 **a**

i $AB = 5$ units

ii $AC = 5$ units

iii $BC = 8$ units

b Isosceles

20

a

- Check if the sides satisfy Pythagoras' Theorem.
- Check if two sides are equal so that the triangle must be isosceles.

b

i $AB = \sqrt{85}$ units

ii $AC = \sqrt{85}$ units

iii $BC = \sqrt{170}$ units

c Yes

21

a

i $PQ = \sqrt{242}$ units

ii $QR = \sqrt{340}$ units

iii $PR = \sqrt{98}$ units

b Yes

22

a

i $MN = \sqrt{8}$ units

ii $NR = \sqrt{50}$ units

b $2 : 5$

23 23.1 units

24

a 50 units^2

b $\sqrt{200}$ units

c $d = 5\sqrt{2}$

25

a $10x$

b $(7x, -2x)$

26

a $3\sqrt{10}$

b $4\sqrt{2}$

c $3\sqrt{10}$

d $4\sqrt{2}$

e $\sqrt{218}$

f $\sqrt{26}$

g Parallelogram

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